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import java.util.HashMap;
import java.util.Map;
import java.util.Scanner;

class Employee {
    int id;
    String name;
    String department;
    double basicSalary;
    double hra;
    double allowance;
    double deductions;

    Employee(int id, String name, String department,
             double basicSalary, double hra,
             double allowance, double deductions) {
        this.id = id;
        this.name = name;
        this.department = department;
        this.basicSalary = basicSalary;
        this.hra = hra;
        this.allowance = allowance;
        this.deductions = deductions;
    }

    double calculateGrossSalary() {
        return basicSalary + hra + allowance;
    }

    double calculateNetSalary() {
        return calculateGrossSalary() - deductions;
    }

    void display() {
        System.out.println("\\nEmployee ID: " + id);
        System.out.println("Name: " + name);
        System.out.println("Department: " + department);
        System.out.printf("Basic Salary: Rs. %.2f%n", basicSalary);
        System.out.printf("HRA: Rs. %.2f%n", hra);
        System.out.printf("Other Allowances: Rs. %.2f%n", allowance);
        System.out.printf("Deductions: Rs. %.2f%n", deductions);
        System.out.printf("Gross Salary: Rs. %.2f%n",
                          calculateGrossSalary());
        System.out.printf("Net Salary: Rs. %.2f%n",
                          calculateNetSalary());
    }
}

public class EmployeeSalaryManagementSystem {
    static Scanner sc = new Scanner(System.in);
    static Map<Integer, Employee> employees = new HashMap<>();

    static void addEmployee() {
        System.out.print("Enter Employee ID: ");
        int id = sc.nextInt();
        sc.nextLine();

        if (employees.containsKey(id)) {
            System.out.println("Employee ID already exists!");
            return;
        }

        System.out.print("Enter Employee Name: ");
        String name = sc.nextLine();

        System.out.print("Enter Department: ");
        String department = sc.nextLine();

        System.out.print("Enter Basic Salary: ");
        double basic = sc.nextDouble();

        System.out.print("Enter HRA: ");
        double hra = sc.nextDouble();

        System.out.print("Enter Other Allowances: ");
        double allowance = sc.nextDouble();

        System.out.print("Enter Deductions: ");
        double deductions = sc.nextDouble();

        if (basic < 0 || hra < 0 || allowance < 0 ||
            deductions < 0 ||
            deductions > basic + hra + allowance) {
            System.out.println("Invalid salary details!");
            return;
        }

        Employee emp = new Employee(
            id, name, department, basic, hra, allowance, deductions
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        );
        employees.put(id, emp);
        System.out.println("Employee added successfully!");
    }

    static void displayEmployees() {
        if (employees.isEmpty()) {
            System.out.println("No employee records found.");
            return;
        }

        for (Employee emp : employees.values()) {
            emp.display();
            System.out.println("----------------------------");
        }
    }

    static void searchEmployee() {
        System.out.print("Enter Employee ID: ");
        int id = sc.nextInt();
        Employee emp = employees.get(id);

        if (emp != null) {
            emp.display();
        } else {
            System.out.println("Employee not found!");
        }
    }

    static void deleteEmployee() {
        System.out.print("Enter Employee ID to delete: ");
        int id = sc.nextInt();

        if (employees.remove(id) != null) {
            System.out.println("Employee deleted successfully!");
        } else {
            System.out.println("Employee not found!");
        }
    }

    public static void main(String[] args) {
        while (true) {
            System.out.println("\\n================================");
            System.out.println(" EMPLOYEE SALARY MANAGEMENT SYSTEM");
            System.out.println("================================");
            System.out.println("1. Add Employee");
            System.out.println("2. Display All Employees");
            System.out.println("3. Search Employee");
            System.out.println("4. Delete Employee");
            System.out.println("5. Exit");
            System.out.print("Enter your choice: ");

            int choice = sc.nextInt();

            switch (choice) {
                case 1:
                    addEmployee();
                    break;
                case 2:
                    displayEmployees();
                    break;
                case 3:
                    searchEmployee();
                    break;
                case 4:
                    deleteEmployee();
                    break;
                case 5:
                    System.out.println("Thank you!");
                    sc.close();
                    return;
                default:
                    System.out.println("Invalid choice!");
            }
        }
    }
}
```